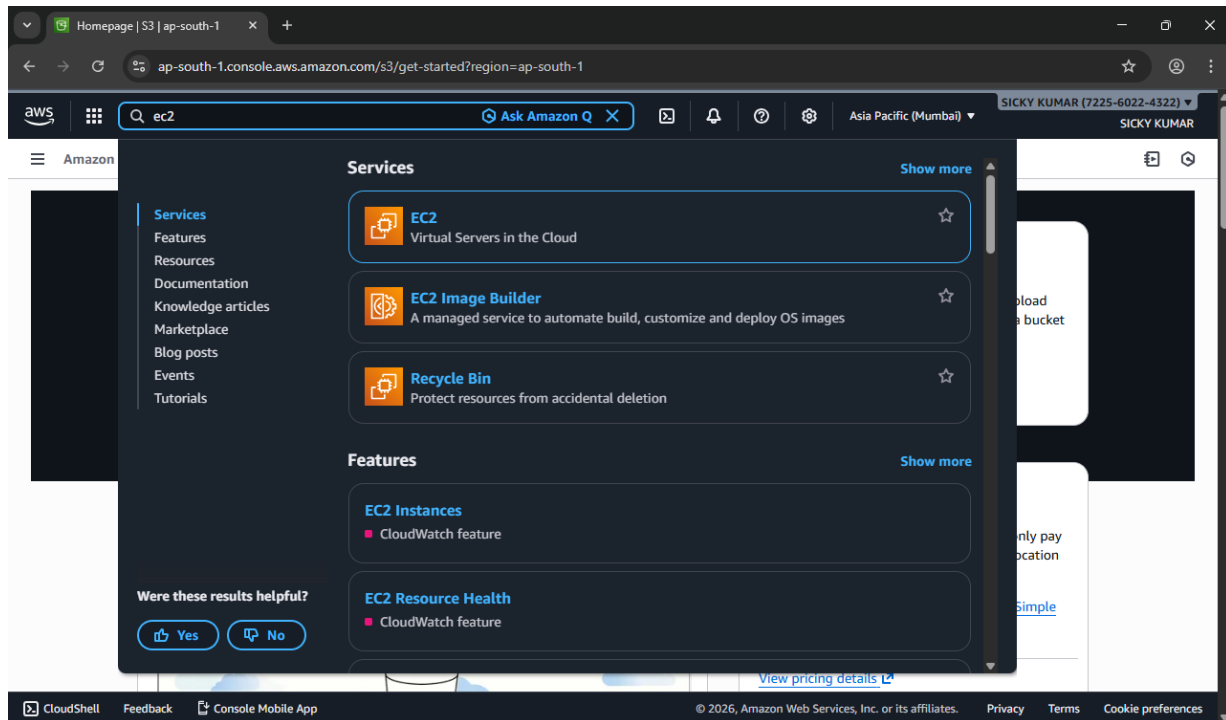
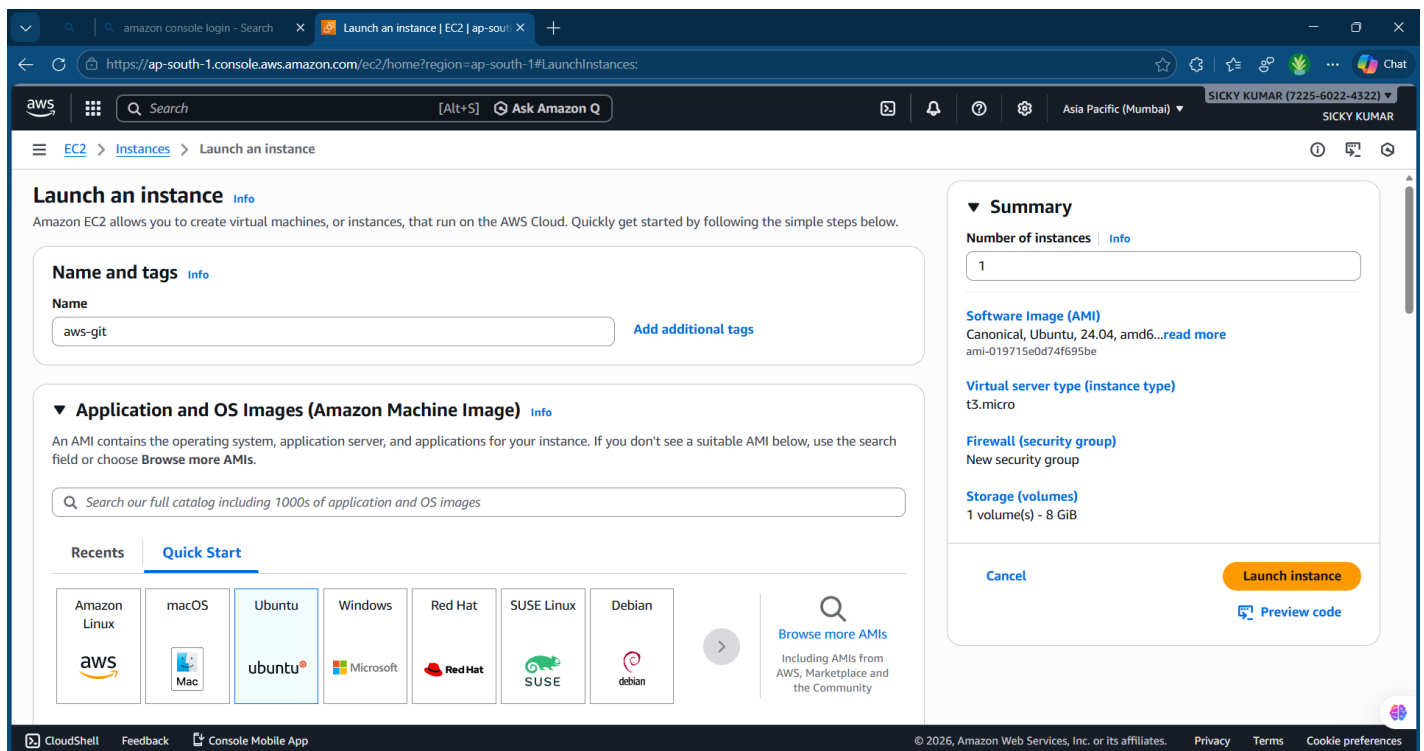


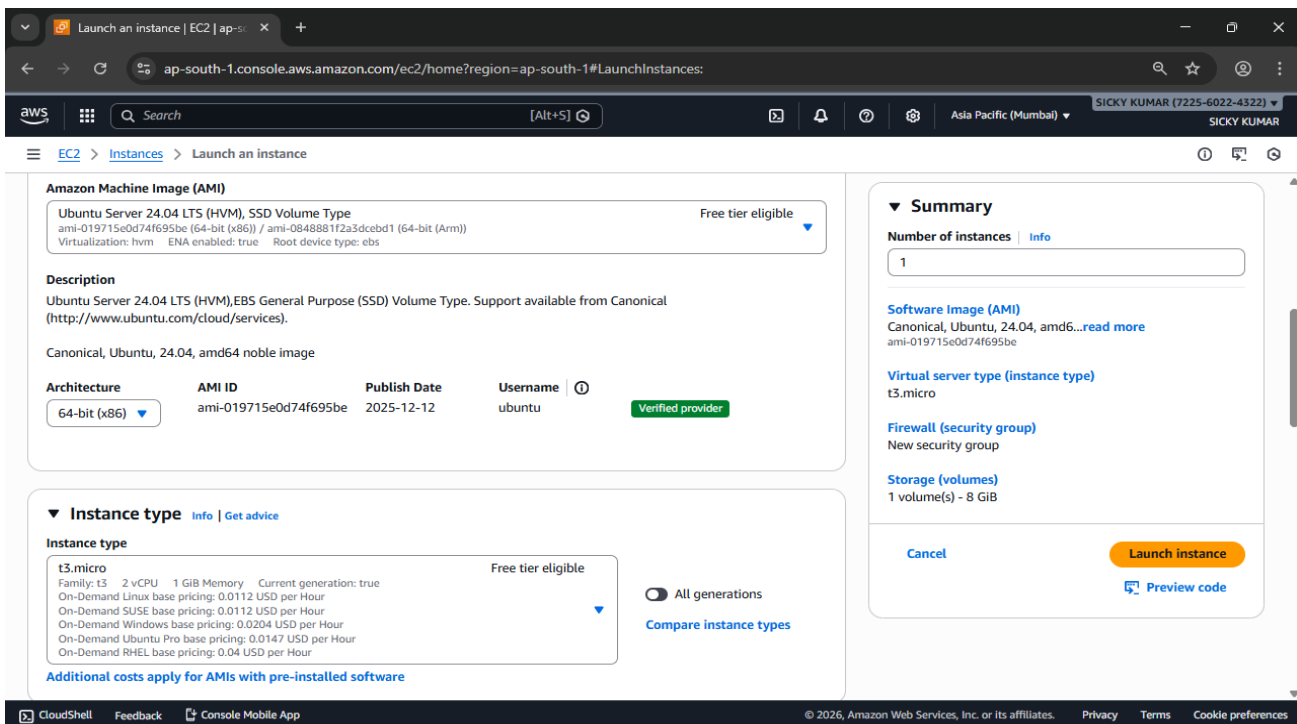
Assignment: 09**Problem Statement:** Deploy a project from GitHub to EC2.**Step 1: Open EC2 Service**

Log in to the AWS Management Console. Search for **EC2** in the top search bar and open the **Amazon EC2** service. This service allows you to create and manage virtual servers

**Step 2: Launch a New EC2 Instance**

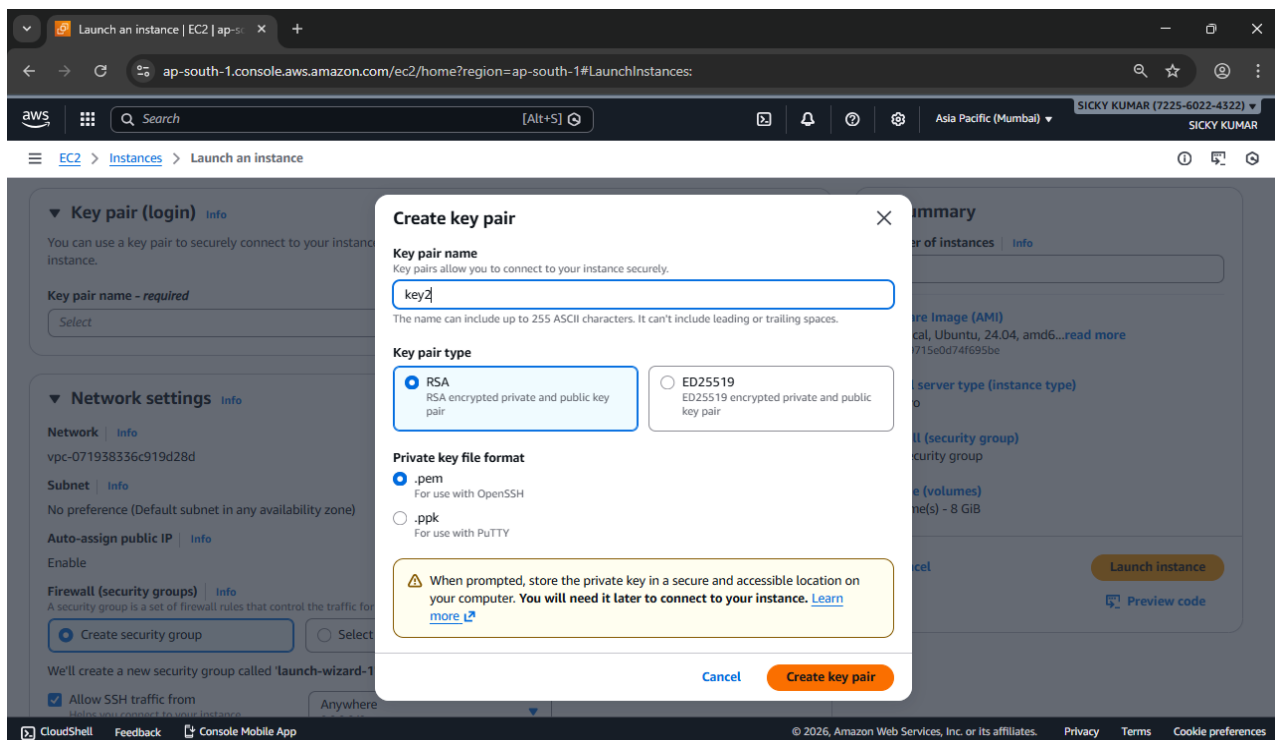
Click **Launch Instance**. Provide a name for the instance (Ec2-Instance). Choose an **Amazon Machine Image (AMI)** such as **Amazon Ubuntu**. Select an **Instance Type** (t3.micro, free-tier eligible).





Step 3: Create or Select a Key Pair

Under **Key pair**, select an existing key pair or create a new one. Download the key pair file (key2.pem) and keep it safe. This key is required to connect to the EC2 instance securely.



Step 4: Configure Security Group

Create a new security group and allow the following inbound rules:

- **SSH (Port 22)** – for remote login
- **HTTP (Port 80)** – for website access Source can be set to **Anywhere (0.0.0.0/0)**.

Launch an instance | EC2 | ap-south-1

ap-south-1.console.aws.amazon.com/ec2/home?region=ap-south-1#LaunchInstances:

Firewall (security groups) [Info](#)

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

☒ Create new security group ☐ Select existing security group

We'll create a new security group called 'launch-wizard-1' with the following rules:

☒ Allow SSH traffic from

☒ Allow HTTPS traffic from the internet
To set up an endpoint, for example when creating a web server

☒ Allow HTTP traffic from the internet
To set up an endpoint, for example when creating a web server

Rules with source of 0.0.0.0/0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only.

▼ Configure storage [Info](#) [Advanced](#)

1x GiB Root volume, 3000 IOPS, Not encrypted

[Add new volume](#)

The selected AMI contains instance store volumes, however the instance does not allow any instance store volumes. None of the instance store volumes from the AMI will be accessible from the instance

▼ Summary

Number of instances

Software Image (AMI)
Canonical, Ubuntu, 24.04, amd64...[read more](#)
ami-019715e0d74f695be

Virtual server type (instance type)
t3.micro

Firewall (security group)
New security group

Storage (volumes)
1 volume(s) - 8 GiB

[Cancel](#) [Launch instance](#) [Preview code](#)

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Step 5: Launch the Instance

Review all settings and click **Launch Instance**.

Wait until the instance state changes to **Running**.

Launch an instance | EC2 | ap-south-1

ap-south-1.console.aws.amazon.com/ec2/home?region=ap-south-1#LaunchInstances:

EC2 > Instances > Launch an instance

Success
Successfully initiated launch of instance (i-0e7f8631005bfee04)

[Launch log](#)

Next Steps

Step 6: Edit Inbound Rules

Go to **Inbound Rules** section and Click on **Add Rules**. then select **Custom TCP** And set PORT range to **4000** and also select source to **Anywhere**.

amazon console login - Search x Launch an instance | EC2 | ap-south-1 x ModifyInboundSecurityGroupRules x

https://ap-south-1.console.aws.amazon.com/ec2/home?region=ap-south-1#ModifyInboundSecurityGroupRules?securityGroupId=sg-09c03c4d5a3530e1d

EC2 > Security Groups > sg-09c03c4d5a3530e1d - launch-wizard-2 > Edit inbound rules

Edit inbound rules [Info](#)

Inbound rules control the incoming traffic that's allowed to reach the instance.

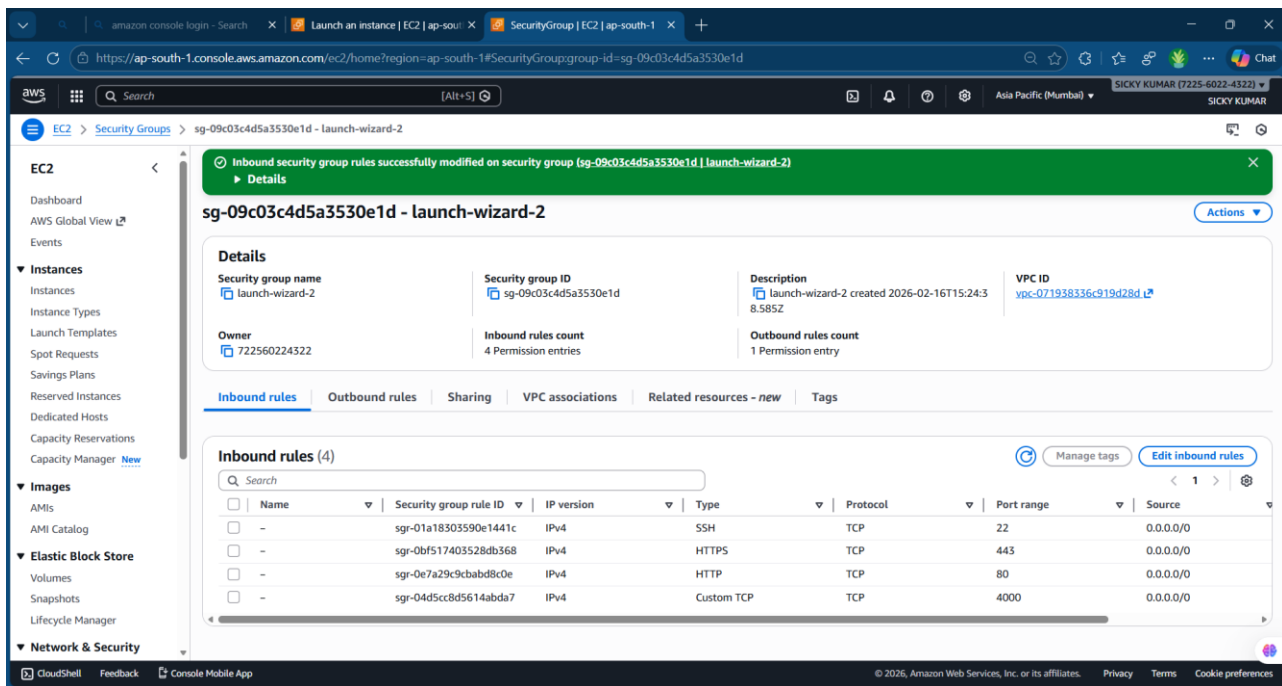
Security group rule ID	Type	Protocol	Port range	Source	Description - optional	
sgr-01a18303590e1441c	SSH	TCP	22	Custom	0.0.0.0/0	Delete
sgr-0bf517403528db368	HTTPS	TCP	443	Custom	0.0.0.0/0	Delete
sgr-0e7a29c9cbadb8c0e	HTTP	TCP	80	Custom	0.0.0.0/0	Delete
-	Custom TCP	TCP	4000	Anywhere...	0.0.0.0/0	Delete

[Add rule](#)

Rules with source of 0.0.0.0/0 or ::/0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only.

[Cancel](#) [Preview changes](#) [Save rules](#)

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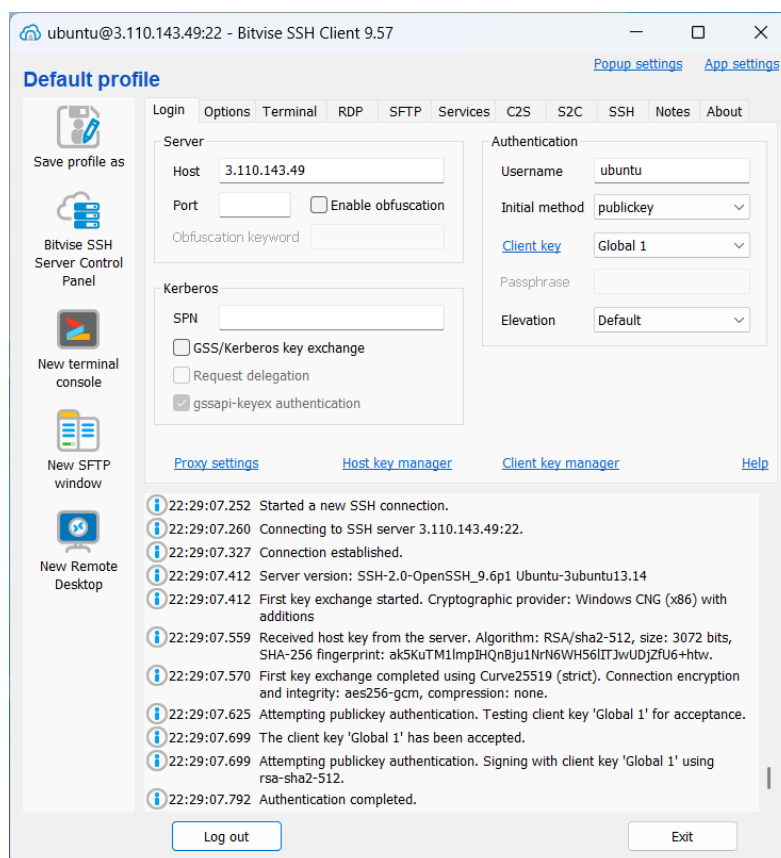


Step 7: Connect EC2 using Bitwise

Open **Bitwise SSH Client** → In **Login** tab enter:

- Host: Public IPv4 address
- Port: 22
- Username: ubuntu

Go to **Authentication** → **Publickey** → **Client key manager** → Import .pem file → Select the key → Click **Login** → Accept host key.



Step 8: Install and Configure Nginx Web Server

After connecting to the EC2 instance, install and start **Nginx**, which is a lightweight and high-performance web server.

1. Update the system packages:

```
sudo apt upgrade -y
```

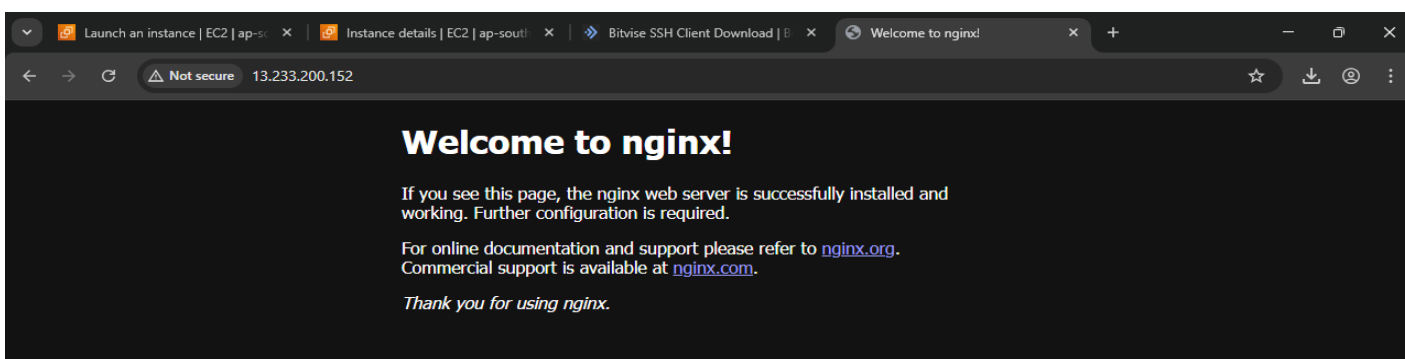
2. Install Nginx:

```
sudo apt install nginx -y
```

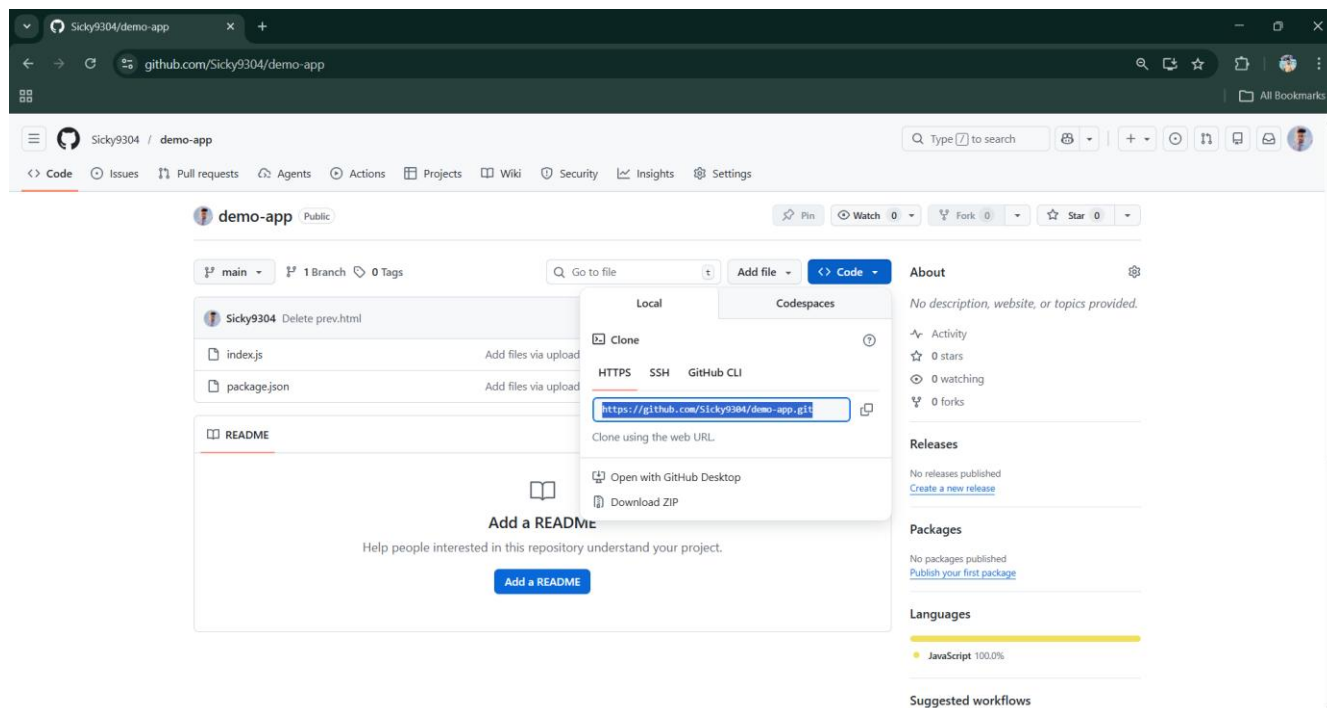
3. Install deb.nodesource setup_16.x

```
ubuntu@ip-172-31-37-117:~$ sudo apt-get install nginx
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
nginx is already the newest version (1.24.0-2ubuntu7.6).
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
ubuntu@ip-172-31-37-117:~$ nginx -v
nginx version: nginx/1.24.0 (Ubuntu)
ubuntu@ip-172-31-37-117:~$ curl -sL https://deb.nodesource.com/setup_16.x | sudo -E bash -
```

4. Verify Nginx is running by opening the EC2 Public IPv4 address in a browser



Step 9: Go to your GitHub repository, open the main page of the project, and click on the green “Code” button. Then copy the HTTPS clone link from the dropdown.



Step 10: Clone Repository and Run the Application

Open terminal in EC2 → Run git clone <repository-link> → Go to project folder using cd demo-app → Install dependencies using npm install → Start the server using node index.js.

```
ubuntu@ip-172-31-37-117:~$ git clone https://github.com/Sicky9304/demo-app.git
Cloning into 'demo-app'...
remote: Enumerating objects: 13, done.
remote: Counting objects: 100% (13/13), done.
remote: Compressing objects: 100% (13/13), done.
remote: Total 13 (delta 5), reused 0 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (13/13), 4.48 KiB | 2.24 MiB/s, done.
Resolving deltas: 100% (5/5), done.
ubuntu@ip-172-31-37-117:~$ cd demo-app
ubuntu@ip-172-31-37-117:~/demo-app$ pwd
/home/ubuntu/demo-app
ubuntu@ip-172-31-37-117:~/demo-app$ ls
index.js  package.json
ubuntu@ip-172-31-37-117:~/demo-app$ npm install

added 224 packages, and audited 225 packages in 13s

26 packages are looking for funding
  run `npm fund` for details

1 low severity vulnerability

Some issues need review, and may require choosing
a different dependency.

Run `npm audit` for details.
ubuntu@ip-172-31-37-117:~/demo-app$ node index.js
Started server
```

Step11: Now we go to the IP 3.110.220.236:4000 to check our hosted project



Step12: Now we go to the IP 3.110.220.236:4000\courses to check our hosted project

